

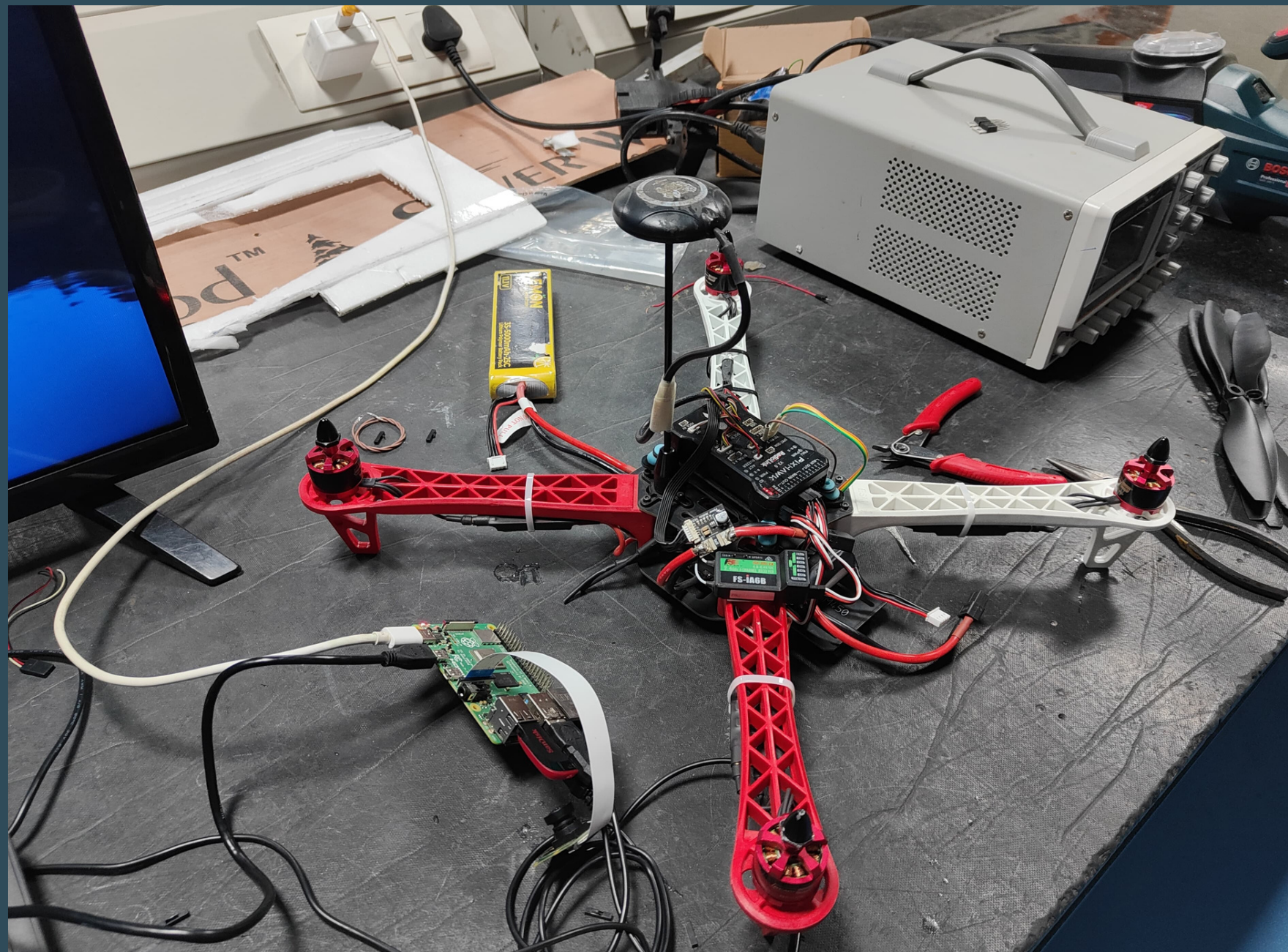




Enabling Autonomous Drone Flight for Delivery of Medical Supplies



Prototype for Autonomous Drone Delivery System



Challenges of Medical Logistics



Remote Areas



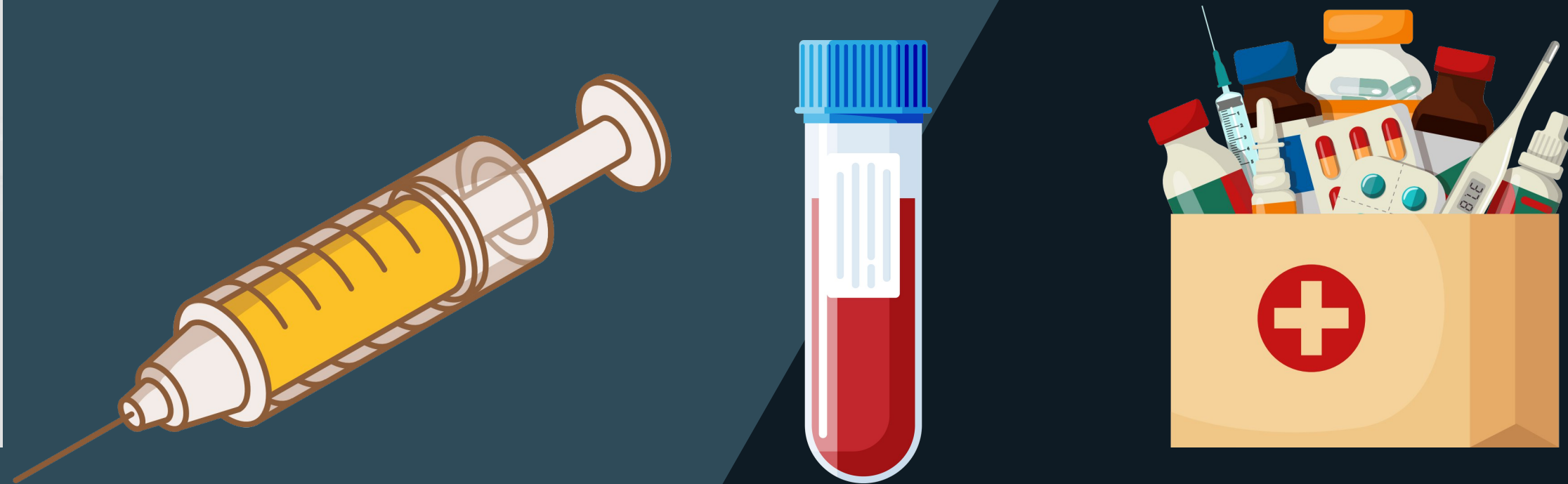
Underserved Regions

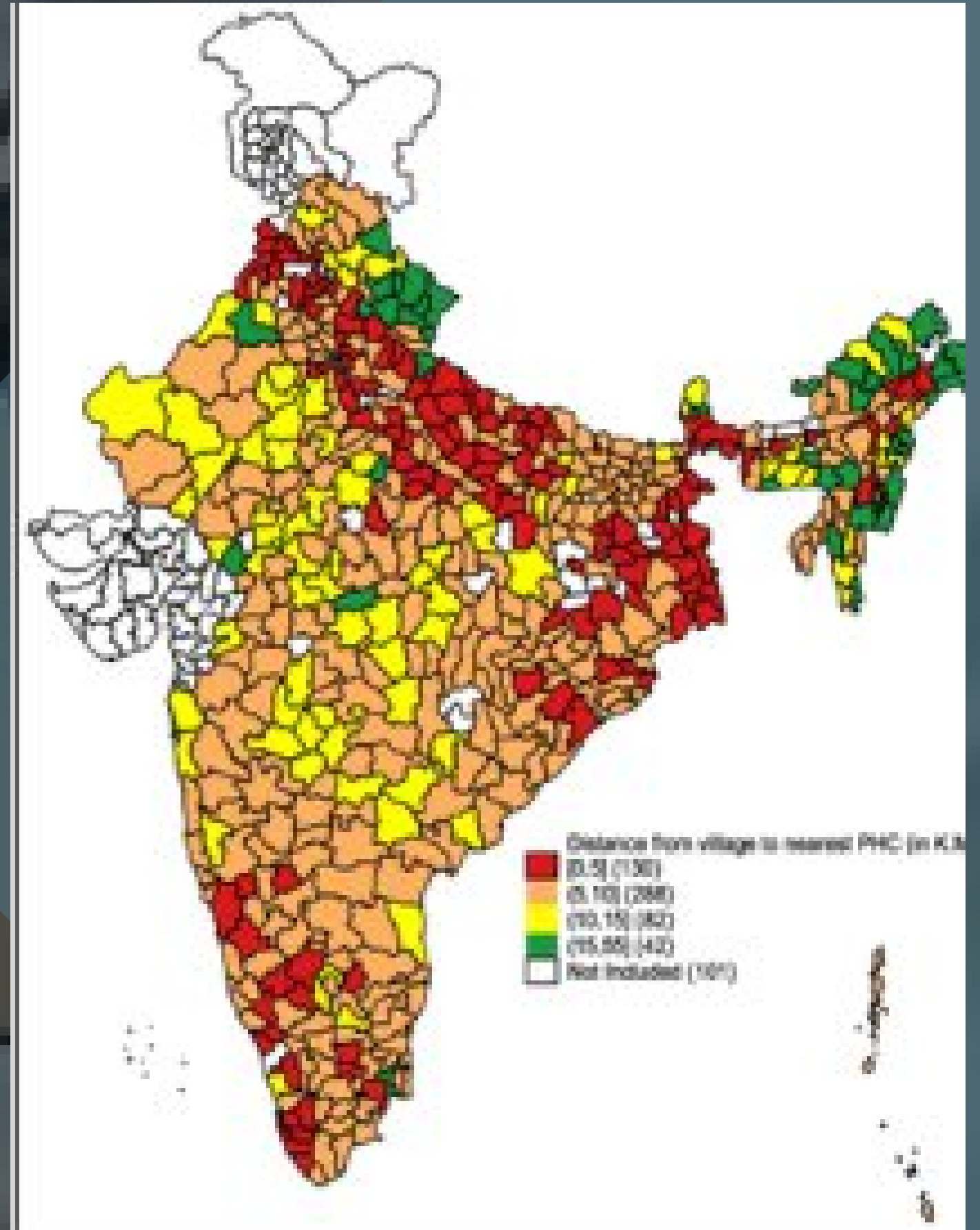
Goal of the Project

- Timely
- Reliable
- Automated



Temperature Controlled Carrier Box







Relevance & Impact of the Solution





Traditional Medical Transport & Delivery Systems

delayed

time-intensive

busy roads



prone to cold chain failures

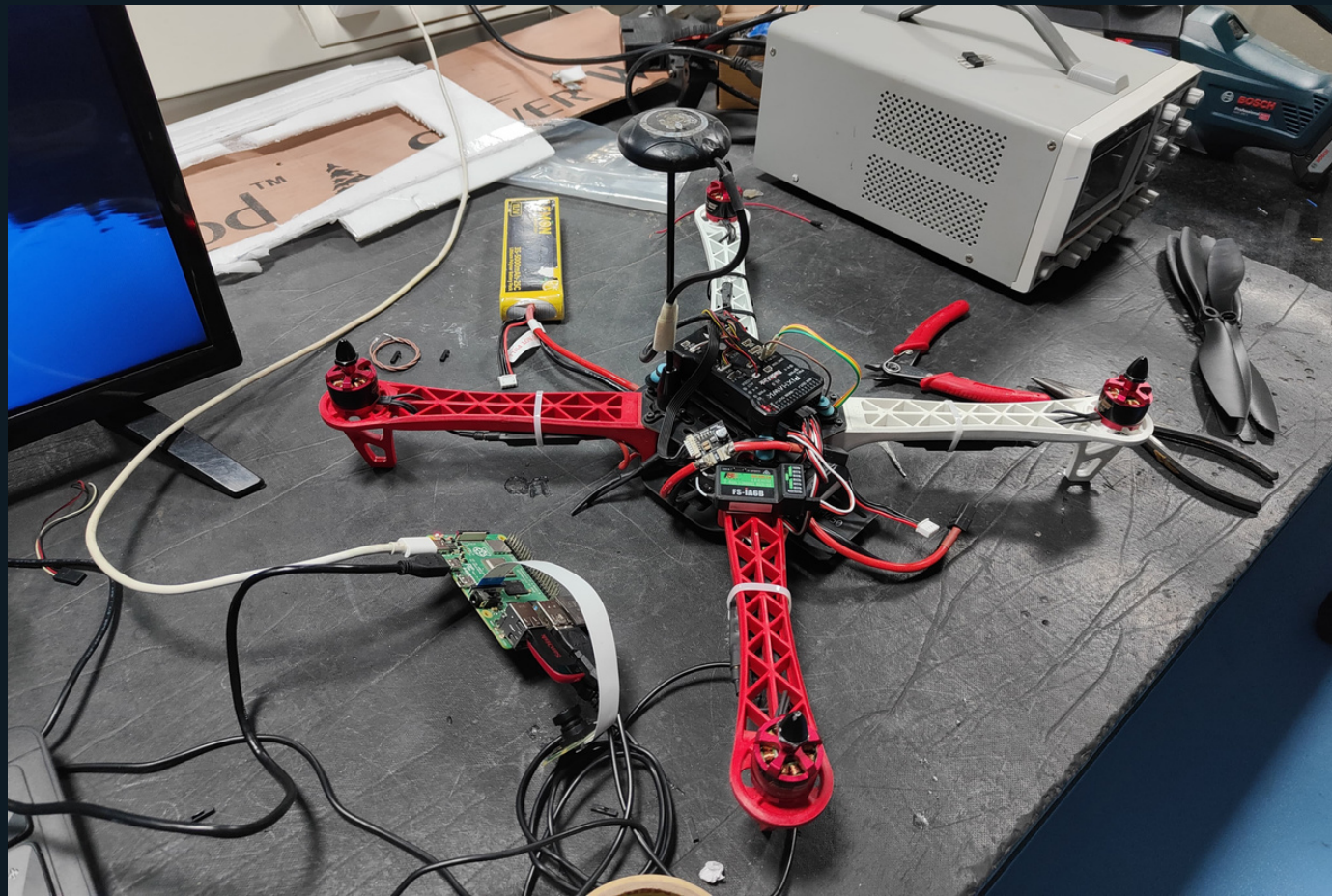
require human resource

Our Autonomous Drone Framework

Fast

Scalable

Reliable



Key Functional Features

Designed for thermally insulated vaccine carrier



Video Telemetry and GPS tracking
Range: max 25 km



Onboard Companion Computer



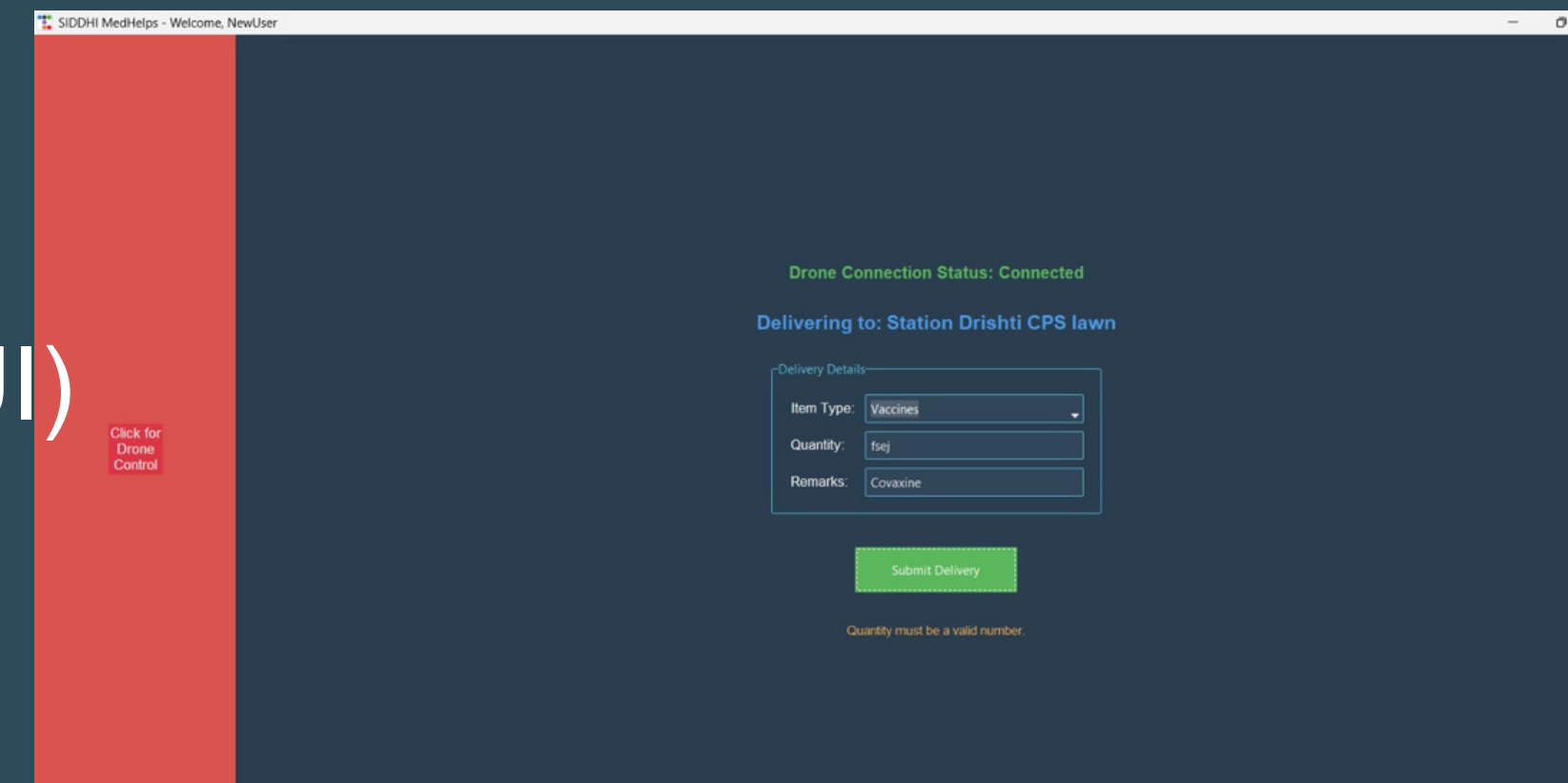
Key Functional Features

Precision Landing (ArUco Markers)

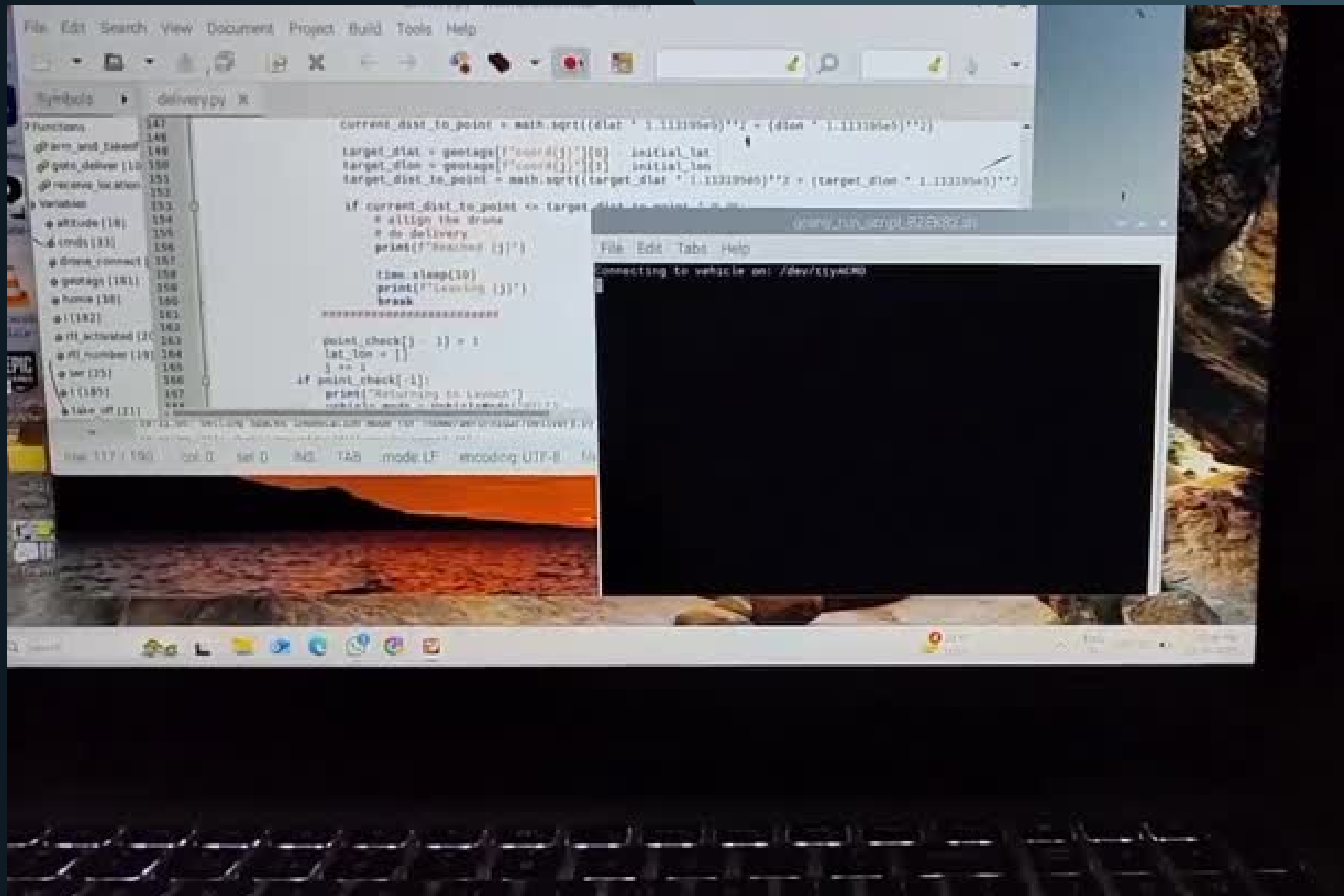


FailSafe Mechanisms

Interactive Graphical User Interface (GUI)



Autonomous Flight Workflow



Manual Takeoff
(acc to DGCA laws)

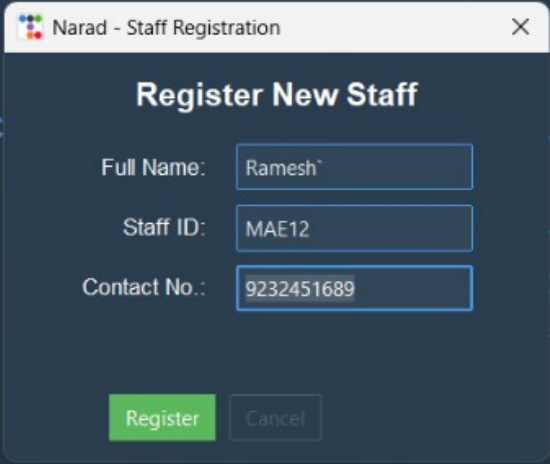


autonomous navigation



autonomous landing at goal

GPS location of Goal set by user via GUI



Narad - Staff Registration

Register New Staff

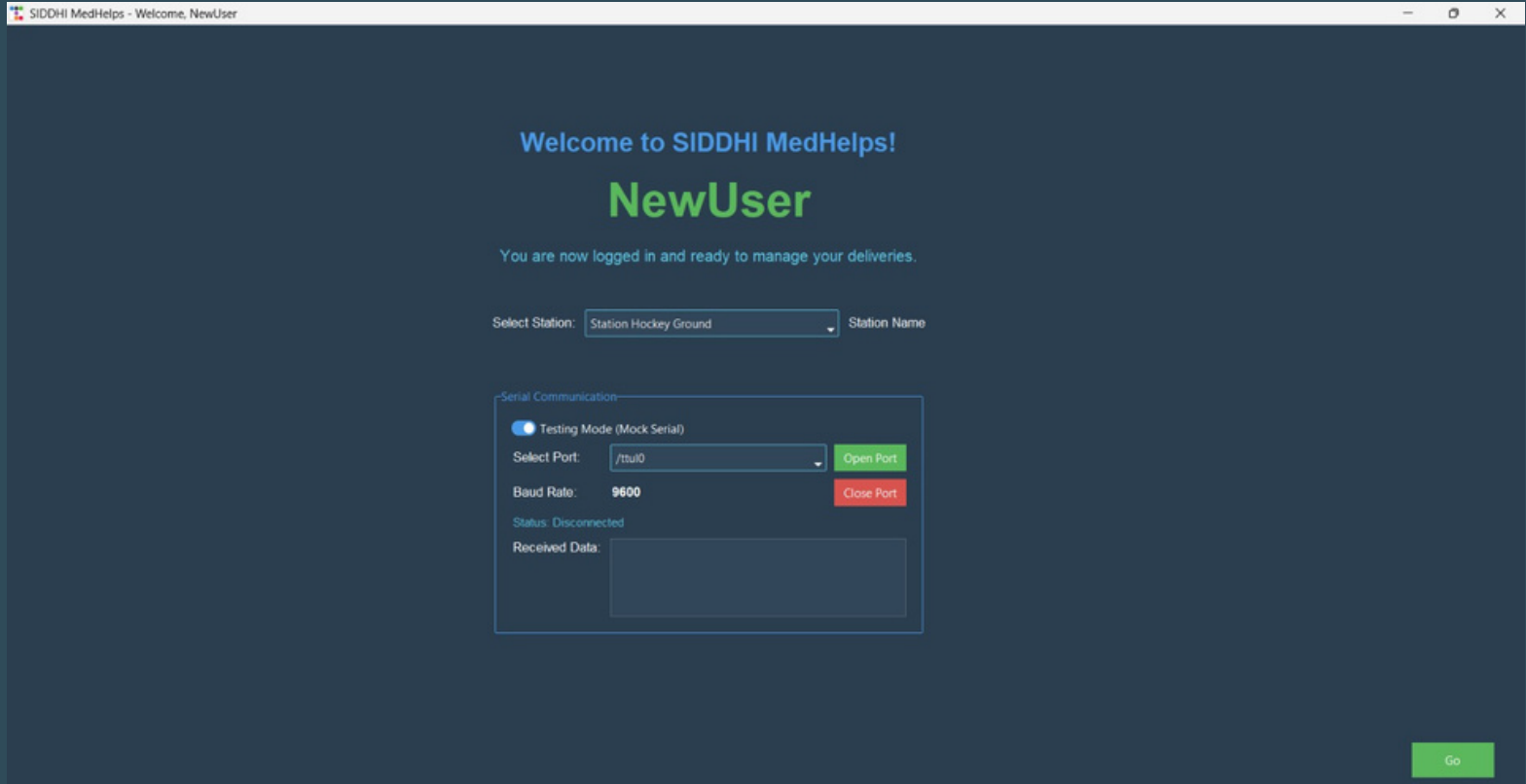
Full Name: Ramesh

Staff ID: MAE12

Contact No.: 9232451689

Register Cancel

Log In



SIDDHI MedHelps - Welcome, NewUser

Welcome to SIDDHI MedHelps!

NewUser

You are now logged in and ready to manage your deliveries.

Select Station: Station Hockey Ground Station Name

Serial Communication

Testing Mode (Mock Serial)

Select Port: /tty0 Open Port

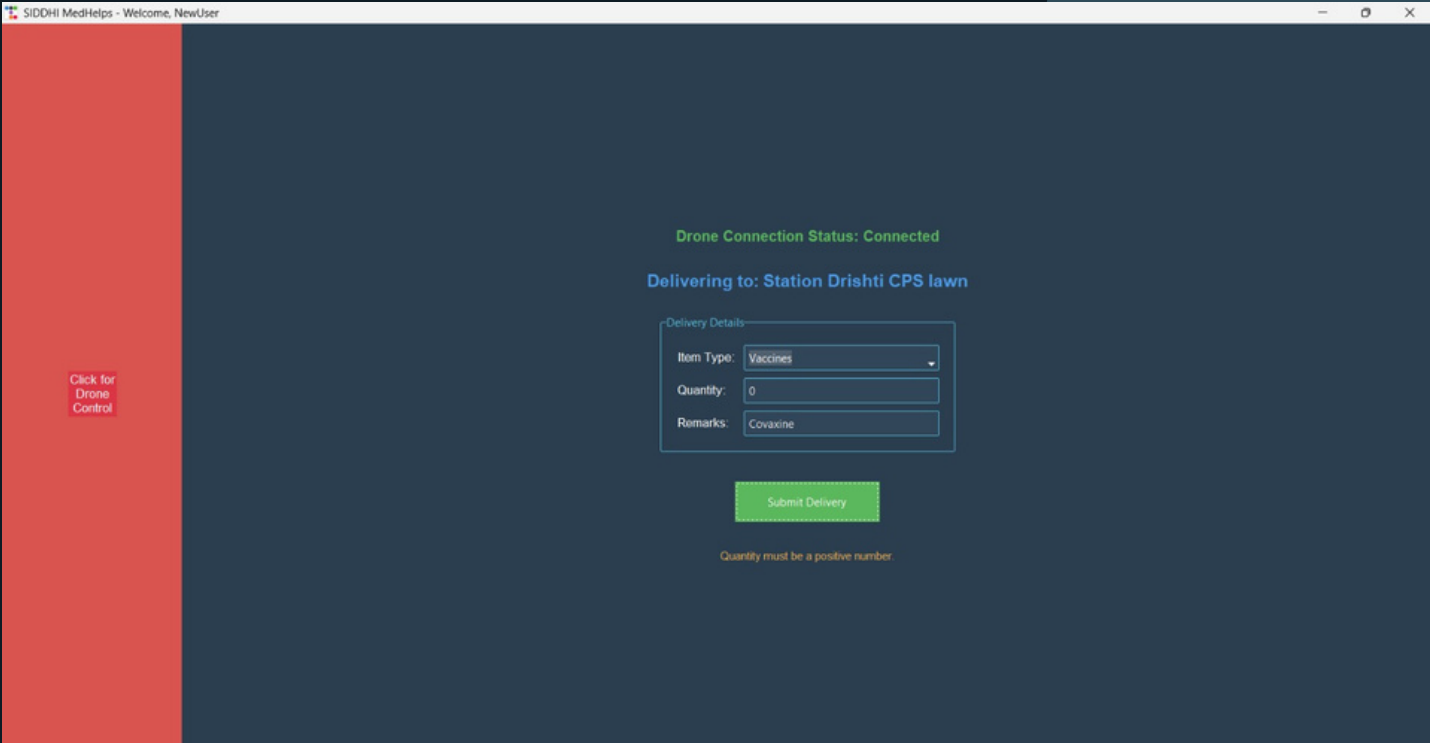
Baud Rate: 9600 Close Port

Status: Disconnected

Received Data:

Go

Select destination and port number



SIDDHI MedHelps - Welcome, NewUser

Drone Connection Status: Connected

Delivering to: Station Drishti CPS lawn

Delivery Details

Item Type: Vaccines

Quantity: 0

Remarks: Covaxine

Submit Delivery

Quantity must be a positive number

Click for Drone Control

Enter payload details and verify location



Use Case Scenarios



High Urgency Remote Missions



Vaccine Deliveries



Vaccine Deliveries



The Drone provides temperature controlled delivery in extreme weather conditions too.

THANKYOU